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Integrating geospatial data and hybrid machine learning models for landslide susceptibility assessment in semi-arid environments

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ABSTRACT

Landslides are destructive natural hazards with severe impacts on human safety, infrastructure, and the environment. Accurate mapping and prediction of landslide-prone areas are essential for effective risk management and mitigation. This study focuses on the Oued Lakhdar watershed, where three hybrid models were developed by combining Random Forest with Multivariate Adaptive Regression Splines, Naïve Bayes Tree, and Boosted Regression Tree. An inventory of 294 landslides, along with twelve conditioning factors such as elevation, slope, and land use, was used for model training and evaluation. The RF-MARS model identified 14.13% of the study area as highly susceptible to landslides, compared to 11.97% with RF-BRT and 8.23% with RF-NBT. In terms of predictive accuracy, RF-MARS achieved the highest AUC (93.66%), followed by RF-BRT (91.96%) and RF-NBT (91.10%), confirming the superior performance of RF-MARS. These findings demonstrate the value of hybrid machine learning approaches in improving landslide susceptibility assessment and supporting natural hazard management.

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Landslide susceptibility; semi-arid zone; High Atlas; machine learning; integrated geospatial data

Introduction

A landslide refers to the downslope displacement of soil and rock materials driven by gravity, which results from the combined effects of geomorphological, geological, meteorological, and hydrological conditions of the terrain (Yusof et al. 2024). These natural phenomena are among the most widespread and common hazards in mountainous and hilly regions, where they can lead to loss of life as well as significant damage to infrastructure and the environment (Kadirhodjaev et al. 2020).

On a global scale, the destruction caused by landslides is critical, with thousands of deaths and hundreds of billions in property damage (Chen and Chen 2021). Sharma et al. (2024) reported that, based on EM-DAT statistics, landslides are responsible for about 17% of natural-disaster-related deaths worldwide (CRED 2022). Several researchers indicate that the socioeconomic impacts of landslides have been exacerbated by unplanned urban development and the effects of climate change (Trinh et al. 2022). In addition, the impact of landslides is often ignored by populations and underestimated due to the lack of systematically collected data (Sim et al. 2022).

In Morocco, these hazards occur mainly in the Rif mountains (Es-smairi et al. 2023). They also affect the High Atlas region, one of the main mountain ranges in North Africa. These areas are particularly vulnerable because of their complex geology, and heavy rainfall and storms can trigger landslides (Ait Naceur et al. 2022).

To reduce damage from landslide disasters in a region, it is critical to determine susceptibility to this hazard (Kim et al. 2018). Landslide susceptibility refers to the probability of a landslide occurring in an area under the influence of interactions between local terrain conditions (Reichenbach et al. 2018). It can

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be determined by assuming that future landslides will occur under conditions identical to those of previous events. Precise identification of areas prone to these phenomena is crucial to reducing the associated risk (Azarafza et al. 2021). The occurrence of landslides is a multifaceted process controlled by geo-environmental conditions such as geological structure, slope gradient, lithological features, elevation, rainfall, and human interventions. Consequently, the primary step in evaluating susceptibility is the collection of information on landslide events and related conditioning parameters (Kadirhodjaev et al. 2020).

To assess landslide susceptibility, several analytical methods have been proposed. Some are knowledge-based, while others are data-driven (Steger et al. 2021; Panahi et al. 2022). Knowledge-based methods, also known as qualitative methods, rely on the expertise and experience of the analyst to assess the risk factors that may influence the likelihood of a landslide (Nwazelibe et al. 2023). The advantage of these methods is that they are easy to implement. However, one drawback is their subjectivity (Li et al. 2022). In addition, these methods cannot be applied in areas where there is no expert knowledge of landslides, which limits their use at present (Hong 2023).

In contrast to qualitative methods, quantitative methods are considered objective. They are based on statistical analysis of data using mathematical models and various algorithms to establish the relationship between observed landslides and causative factors. Another advantage of these methods is the structuring of assessment processes, which enables automation and standardisation of data management. Among the most popular quantitative methods are frequency ratio (FR) (Panchal and Shrivastava 2021; Alsabhan et al. 2022; Biswas et al. 2023) weight of evidence (WoE) (Cao et al. 2021; Alsabhan et al. 2022), and Information Value (IV) (Akinci and Yavuz Ozalp 2021; Ait Naceur et al. 2021; Singh et al. 2021). These techniques have been widely utilised to evaluate landslide susceptibility since they are straightforward to apply and comprehend (Jiao et al. 2019).

In recent years, with the development of informatics tools, many machine learning algorithms have been used for predicting landslide susceptibility (Zhao et al. 2020a). These tools have attracted the attention of researchers worldwide because of their flexibility, good performance, and ability to handle large datasets (Khosravi et al. 2020; Ngo et al. 2020). The results produced are relatively more accurate, offering a firmer basis for landslide risk assessment and mitigation in the future (Pourghasemi et al. 2020).

In the last ten years, machine learning approaches have played a major role in improving the mapping-based evaluation of landslide vulnerability (Kainthura and Sharma 2022). The limitations of traditional statistical approaches, which include making assumptions before modelling, can be circumvented using machine learning models (Dikshit et al. 2021). For assessing landslide vulnerability, some popular machine learning models include logistic regression (LR) (Shang et al. 2024; Zhao et al. 2024); Radial Basis Function Network (RBFN) (Ait Naceur et al. 2022; Luu et al. 2023) and naive Bayes (NB) (Nhu et al. 2020; Shang et al. 2024; Bhargavi and Jyothi 2009) and support vector machine (SVM) (Zhang et al. 2023; Xu et al. 2024) Artificial Neural Network (ANN) (Moayedi et al. 2024).

However, these models still have several shortcomings that affect the prediction performance of the individual models. For example, there is a risk of under-ajustement when training data is limited, which could result in incorrect model development if the model is used alone (Kainthura and Sharma 2022).

To solve this issue, ensemble learning models have been developed in the machine learning field. The principle of these techniques is to generate and combine a group of basic models using different methods, such as boosting or bagging, to accomplish the learning task. Among these models, the most widely used are: Random Forest (RF) (França Pereira et al. 2023; Daviran et al. 2023); Rotation Forest (RF) (Fang et al. 2021; Shen et al. 2023); Gradient Boosting Decision Trees (GBDT) (Tang et al. 2023; Dyer et al. 2024); and AdaBoost (AB) (Wen et al. 2023; Saha et al. 2023; Ait Naceur et al. 2024).

Ensemble techniques enable us to acquire a better knowledge of the data and rules of several models, thus reducing variance and bias. They can also help reduce overfitting problems when data has a wide variety of dimensions or are unbalanced (McCoy et al. 2021). In addition, computational performance is not affected by missing values in the dataset (Aleryani et al. 2020). However, model ensembles are not always preferable. A limitation of bagging is its dependence on the mean prediction of the subset of trees, instead of delivering precise values in classification or regression tasks (Aria 2021). On the other hand, overlearning is possible in boosting if parameters are not properly defined (Mehta et al. 2019). Saha et al. (2021) concludes that the capacity of each individual model has certain limits.

Many studies have demonstrated the hybridisation of models for landslide susceptibility prediction worldwide, generally improving model performance and prediction accuracy (Kainthura and Sharma 2022; Lv et al. 2022). To the best of our knowledge, no previous study has combined and comparatively evaluated Random Forest (Zhao et al. 2020b; Badapalli et al. 2025) with Multivariate Adaptive Regression Splines (Chu et al. 2019), Boosted Regression Trees (Youssef et al. 2016), and Naïve Bayes Tree (Hu et al. 2021; Arabameri et al. 2021) for landslide susceptibility prediction.

The aim of this paper is therefore to apply this novel hybrid combination in the Oued Lakhdar watershed, located in the northern part of the central High Atlas, a region particularly prone to landslides due to its complex geological and topographical features. By integrating RF with MARS, BRT, and NBT, this study not only leverages the strengths of each algorithm but also provides a comprehensive comparison of their predictive performance in a semi-arid mountainous environment, offering new insights for local hazard assessment and risk management.

Materials and methods

Study area

The Oued Lakhdar watershed covers an area of approximately 1,277 km² upstream of the Hassan I Dam and is located in the western part of Azilal Province. It extends between longitudes 6°09′04″ W and 6°49′26″ W, and latitudes 31°29′44″ *N* and 31°58′28″ *N* (Figure 1). This territory is characterised by a hilly and mountainous relief and is home to several tourist sites that form part of the Mgoun Geopark. The Lakhder river, which rises in the Zawyat Olmzi region, is one of the main tributaries of the Oued Oum Errabia. Topographically, the maximum elevation of the basin is 3997 metres, with more than 23% of the land above 2500 metres. The basin has an average slope of approximately 28.61°, with local gradients reaching up to 75°. From a geological perspective, the Oued Lakhdar watershed is underlain by Hercynian

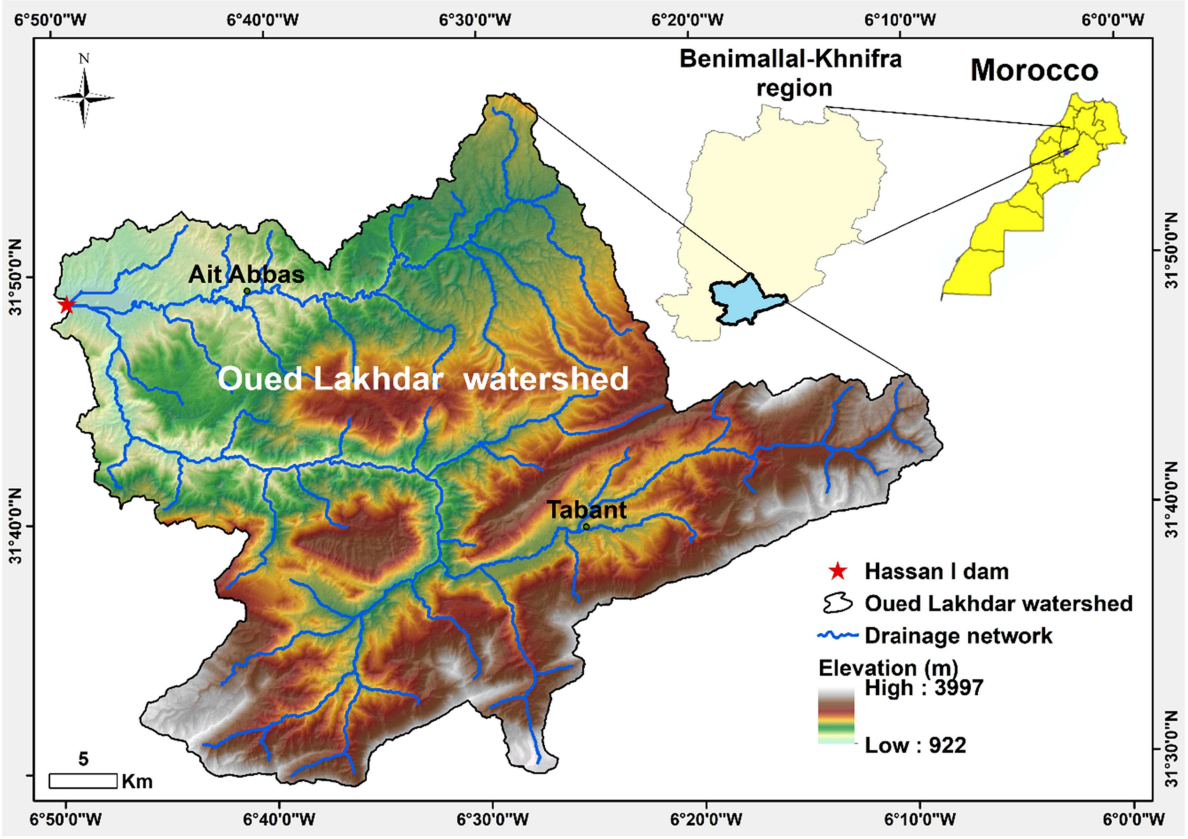


Figure 1. Geographic location of the study area.

basement formations, which include Ordovician units mainly composed of dolomite and marl, locally intruded by doleritic bodies. These are overlain by Triassic red sandstones and marls, with locally developed Late Triassic basaltic flows. The Lias is characterised by oncolitic limestones, while the Jurassic formations are dominated by conglomerates and red sandstones. The youngest deposits correspond to Quaternary fluvial terraces. Morphologically, the southern sector of the watershed is characterised by rugged relief and narrow, elongated valleys. Overall, the Oued Lakhdar basin presents a contrasting landscape of high ridges, rolling plateaus, and deeply incised valleys, reflecting significant geomorphological and lithological heterogeneity.

Methodology

The methodological workflow comprised five sequential steps, as illustrated in Figure 2. Firstly, a comprehensive inventory map of historical landslides in the Oued Lakhdar watershed was established. Secondly, twelve causal-factor maps were generated and used to extract data for multicollinearity assessment among variables. Thirdly, the RF model was trained on the dataset to predict landslide occurrences. The predicted probabilities from RF were then used as additional input features for the hybrid models: MARS, BRT, and NBT. This stacking of probabilities allows the hybrid models to leverage the information captured by RF, improving predictive performance while compensating for the limitations of individual models. All models were developed using Python 3.7 and relevant libraries. Fourthly, all models were evaluated using standard accuracy metrics, and their hyperparameters were fine-tuned to optimise predictive performance. Fifthly, final landslide susceptibility maps were generated and classified into five categories (very low, low, moderate, high, and very high) using the Natural Breaks algorithm in ArcGIS 10.5, which identifies natural groupings within the dataset (Ait Naceur et al. 2024).

Data sources

In the present study, the assessment of landslide susceptibility in the Oued Lakhdar watershed was based on a hybrid approach involving several machine learning algorithms. Multiple data sources were used to create thematic maps of causal factors using GIS and remote sensing tools. A 30 m resolution DEM

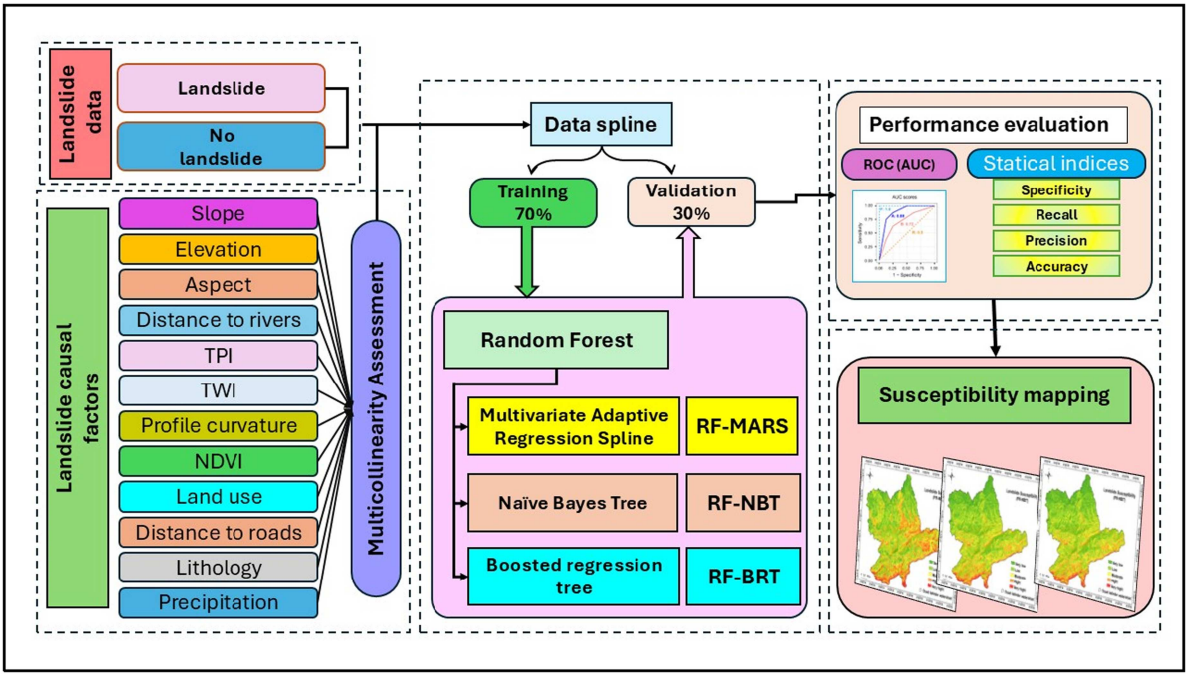


Figure 2. Flowchart of the methodological framework adopted in this study.

(SRTM, 2018) was employed to extract hydrographic, topographic, and geomorphological features, including slope, aspect, curvature, TWI, and TPI. Satellite images from Landsat 8/9 OLI/TIRS (2013–2024), acquired from June to August, were used to calculate vegetation indices (NDVI), ensuring that only permanent vegetation was captured. Land use was also derived from these images. The rainfall map was derived from regional data provided by the Oum Errabia Water Basin Agency (2010–2022). All datasets were reprojected to WGS 84/UTM zone 29 N and harmonised to 30 m resolution. Preprocessing steps, such as raster resampling, calculation of derived indices, and digitisation, were applied to ensure consistency and reproducibility (Table 1).

Historical landslide location

According to Abbas et al. (2024), landslides follow specific geotechnical patterns and do not occur randomly, making landslide inventories essential for understanding the relationship between past events and their controlling factors. In this study, a total of 4 753 landslide pixels within the Oued Lakhdar watershed were delineated using high-resolution Google Earth imagery combined with extensive field surveys, and a roughly equal number of non-landslide pixels were selected. All data were converted to a 30 m raster format to ensure consistency with the other datasets used in the modelling process. For machine learning, the dataset was divided into training and validation subsets. Specifically, 3,327 landslide and 3,327 non-landslide pixels (≈70% of the total) were randomly assigned to the training dataset, while the remaining 1,426 landslide and 1,426 non-landslide pixels (≈30%) were used for validation. Figure 3 illustrates the spatial distribution of these samples, where the red and yellow points represent the centroids of the landslide and non-landslide polygons used for training and validation, respectively. This approach ensures a balanced representation of both classes in the model development and evaluation process. The training and validation datasets were kept strictly independent and consistently applied across all models to ensure fair performance comparison, while additional validation strategies are described in the model evaluation section.

Landslide conditioning factors and preprocessing

Landslides are complex phenomena influenced by the interaction of various factors that can contribute to their triggering. However, no standardised criteria exist for identifying the causal factors of landslides (Panahi et al. 2022). A landslide susceptibility model seeks to establish a relationship between the landslide inventory and the conditioning factors. To achieve high accuracy in predicting vulnerable areas, the selection and preparation of databases of landslide conditioning factors are crucial (Youssef and Pourghasemi 2021). Twelve conditioning factors were selected in this study based on information gathered in the literature and data availability (Fang et al. 2021; Ait Naceur et al. 2022; Tang et al. 2023; Shen et al. 2023; Dyer et al. 2024). These factors include elevation, slope angle, distance to roads, aspect, TPI, lithology, TWI, distance to rivers, Normalised Difference Vegetation Index (NDVI), precipitations, profile curvature, and land use. Categorical conditioning factors, such as aspect, were transformed into numerical

Table 1. Sources and characteristics of the data used in this study.

Category	Factor	Source	Scale/resolution
Topographical	Aspect	DEM	30 m
	Profile curvature	https://earthexplorer.usgs.gov	30 m
	Elevation		30 m
	Slope		30 m
	Topographic Position Index (TPI)		30 m
	Topographic Wetness Index (TWI)		30 m
Meteorological	Rainfall	Oum Errabia basin hydraulic agency	-
Environmental	NDVI	Landsat 8-9 OLI/TIRS https://earthexplorer.usgs.gov	30 m
	Distance to rivers	DEM https://earthexplorer.usgs.gov	30 m
Geological	Lithology	Geologic map (Marrakesh 1/500.000)	1/500.000
Anthropological	Land use	Landsat 8 OLI/TIRS https://earthexplorer.usgs.gov	30 m
	Distance to roads	https://extract.bbbike.org	

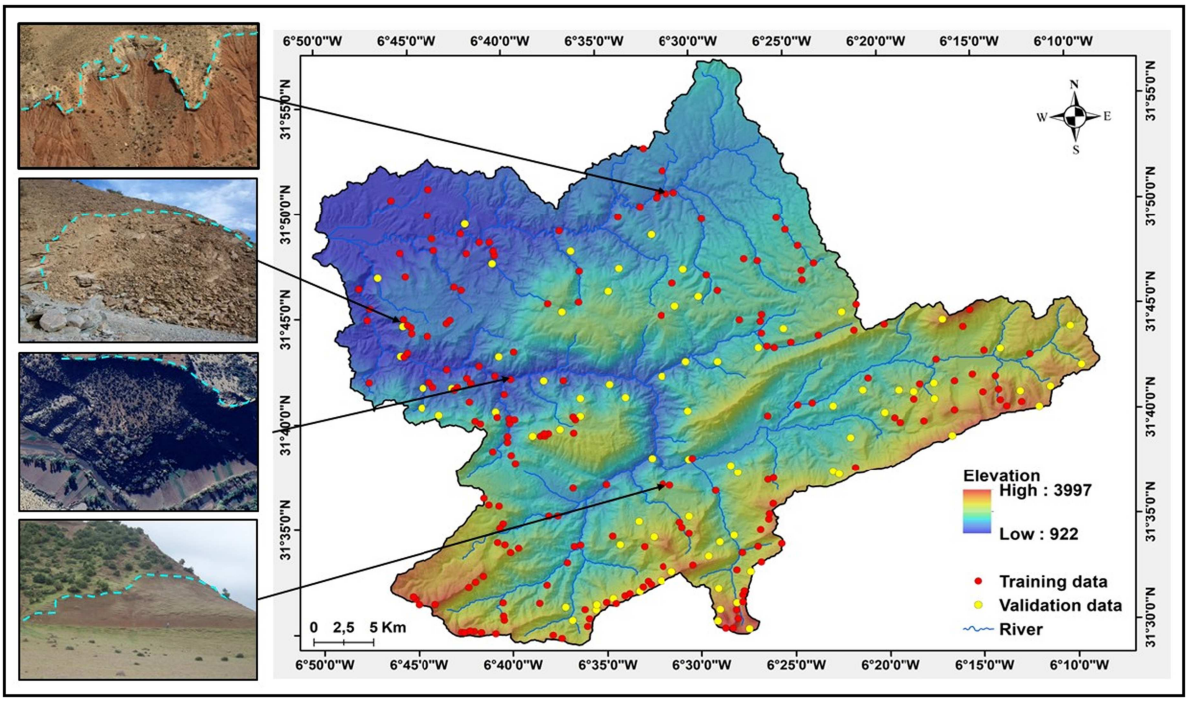


Figure 3. Location map of inventoried historical landslide events in the Oued Lakhdar watershed, with selected examples.

features using one-hot encoding (Zhu et al. 2024). In this method, each category of a variable is represented by a separate binary column, where a value of 1 indicates the presence of the category and 0 indicates its absence. This approach allows the model to process categorical variables without assuming any ordinal relationship between categories.

Slope angle is a crucial factor controlling groundwater flow processes and influencing soil moisture concentration, which is directly linked to landslide occurrence. The greater the slope angle, the greater the probability of landslide occurrence (Youssef and Pourghasemi 2021). The slope factor for the Oued Lakhder catchment area was generated from a DEM and then reclassified into six classes (0-5, 5-10, 10-15, 15-25, 25-35, 35-75°) (Figure 4a).

The utilisation of elevation in landslide susceptibility mapping is common practice in previous studies (Trinh et al. 2022). This factor indirectly influences slope instability and landslides through its effect on vegetation cover and precipitation (Shano et al. 2021). In the Oued Lakhder watershed, terrain heights vary between 922 m and 3 997 m. The corresponding map was generated by dividing the study area into six altitudinal ranges: 922–1500 m, 1500–2000 m, 2000–2500 m, 2500–3000 m, 3 000–3500 m, and 3500–3997 m (Figure 4b).

NDVI is an essential indicator for assessing the quality of vegetation cover, which reflects vegetation–soil interactions, a crucial factor in landslide occurrence (Sharma et al. 2024). This index is calculated from the reflectance of red and near-infra-red bands. In our study, the NDVI map of the Oued Lakhder watershed was derived from Landsat 8 imagery and subsequently reclassified into six classes using the Natural Breaks (Jenks) method to optimise the separation of pixel values. The six classes represent different levels of vegetation density: very low, low, moderate, high, very high, and dense vegetation (Figure 4c).

The lithological features, which include the physical, chemical, mechanical, and hydraulic properties of rocks, have a significant influence on landslide occurrence, making lithology a critical factor for vulnerability assessment (Sulaiman et al. 2019). For this study, the lithological map was created by digitising the lithological units from the geological map of Morocco at a scale of 1:500,000 (Figure 4d). The vector lithological units were then converted into a raster format with a 30 m spatial resolution, consistent with the other datasets used in the modelling process. Each pixel was assigned the lithological class of the underlying polygon, allowing the lithology factor to be directly incorporated as an input variable in the machine learning models.

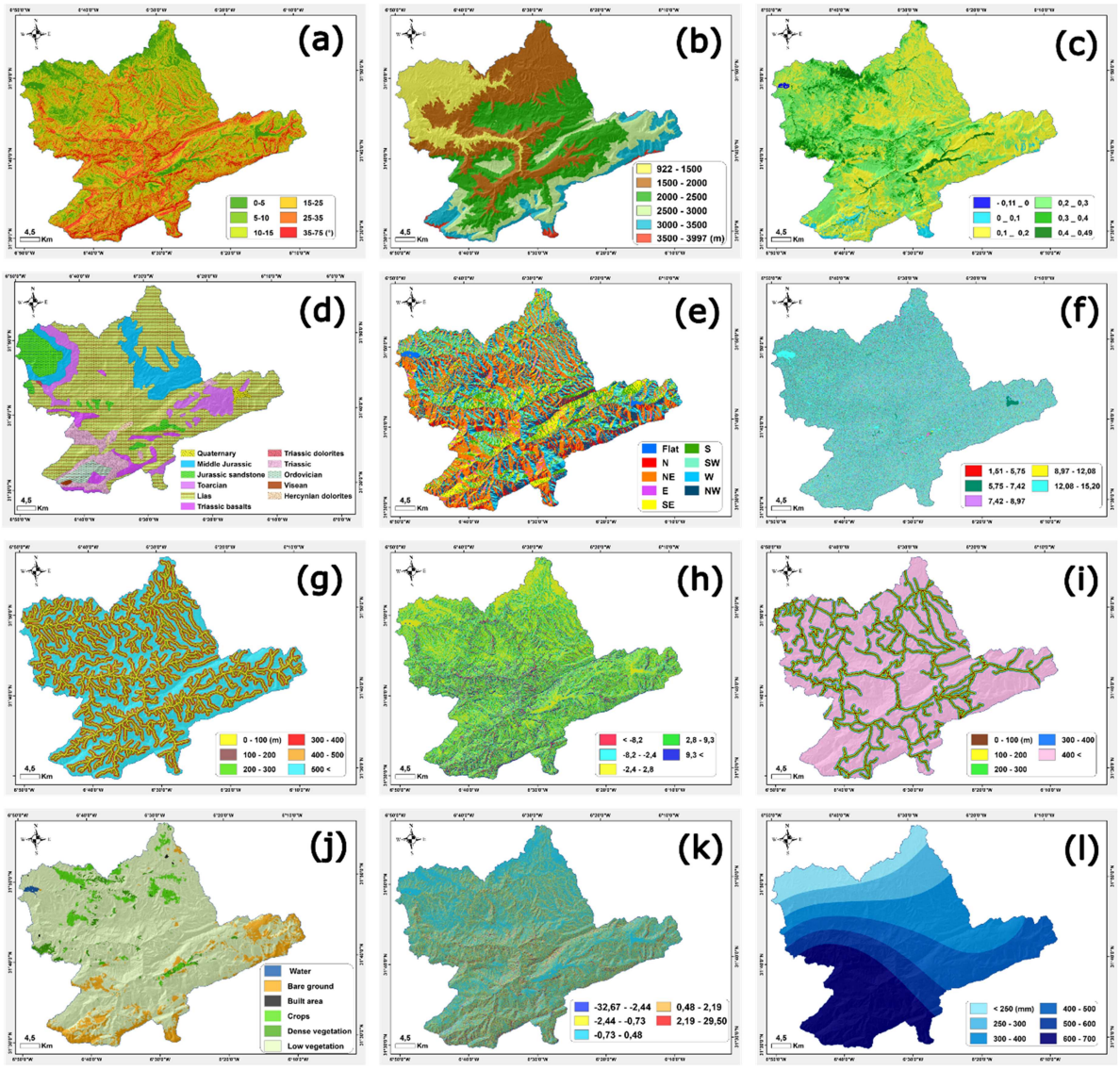


Figure 4. Spatial distribution of landslide conditioning factors in the Oued lakhder Basin((a) Slope, (b) Elevation, (c) NDVI, (d) Lithology, (e) Aspect, (f) TPI, (g) Distance to rivers, (h) TWI, (i) Distance to roads, (j) Land use, (k) Profile curvature, (l) Precipitation).

The slope aspect, or terrain orientation, plays a very important role in the occurrence of landslides by influencing various environmental factors. Sun exposure, humidity, evapotranspiration, and vegetation distribution vary with orientation, affecting slope stability (Ait Naceur et al. 2022). Usually, the south-facing slopes receive more sun, which can lead to faster drying and a different vegetation rate compared to the north-facing slopes. To assess this influence, we generated the aspect map for our study area using data derived from an MNT file. The values of this map have been reclassified into nine classes: flat, N, NE, E, SE, S, SW, W, and NW (Figure 4e).

TPI quantifies the relative position of a cell within the terrain by comparing its elevation to that of the surrounding cells (Chowdhury et al. 2024). Negative TPI values correspond to areas lying below their neighbourhood, such as valleys, while values near zero typically represent flat or uniform slopes. Positive TPI values represent locations above the average of their surroundings, as defined by the neighbourhood (Ait Naceur et al. 2022). TPI values in the Oued Lakhder watershed range from -15 to over 12 and we have classified them into four classes (Figure 4f).

Distance to rivers is a determining factor in the occurrence of landslides (Youssef and Pourghasemi 2021). In fact, areas near watercourses are often more vulnerable due to continuous bank erosion, soil saturation through water infiltration, and increased pore pressure. Water weakens the cohesive forces of soil and rock materials, making slopes more likely to destabilise. The distance-to-river map was generated using the hydrographic network, with buffer zones established at 0–100 m, 100–200 m, 200–300 m, 300–400 m, 400–500 m, and greater than 500 m (Figure 4g).

TWI indicates soil moisture levels and is affected by both the drainage characteristics and precipitation patterns. A high TWI value indicates a relatively large amount of water stored in the soil, which can affect the occurrence of landslides (Irawan et al. 2021). From a DEM we were able to create and reclassify the thematic map of this index for our study area (Figure 4h).

Distance to roads is a factor that has an impact on the incidence of landslides (Al-Najjar and Pradhan 2021). Excavation work during road construction or widening results in the creation of cut slopes, which destabilise the slopes in the vicinity of the road lines. The Wadi Lakhder basin has been divided into six buffer zones in relation to the road line: 0-100 m, 100-200 m, 200-300 m, 300-400 m, and >400 m (Figure 4i).

Concerning land use, it is noted that areas with low vegetation density are more vulnerable to landslides than areas with dense vegetation (Yusof and Pradhan 2014). Through a reclassification of Landsat 8 images, we were able to produce the land use map for the study region (Figure 4j). This map was validated by observations gathered during our field missions. Precipitation is a key natural factor influencing landslide occurrence, as it can reduce slope stability and trigger failures (Grelle et al., 2014). During rainfall infiltration, suction decreases, leading to reduced shear strength and higher landslide risk (Ait Naceur et al. 2022). Using data from the Oum Errabia Basin Hydraulic Agency, we generated a spatial map of the annual mean precipitation for the Oued Lakhdar watershed, interpolated from multiple stations and reclassified into six classes (Figure 4l). The annual precipitation averages were computed over the 2000–2023 period, ensuring a representative long-term climatological baseline for the study area.

Multicollinearity assessment

Multicollinearity assessment is essential to ensure the independence of explanatory variables when assessing susceptibility to landslides. In this study, the multicollinearity of predictive factors was examined using the variance inflation factor (VIF) and tolerance index (TOL), as recommended by several researchers (Merghadi et al. 2020; Namous et al. 2021). A VIF greater than 10 and a tolerance value less than 0.1 generally indicate high multicollinearity between variables, potentially distorting coefficient estimates (Ait Naceur et al. 2024). These indicators were calculated using the following equations:

$$Tolerance = 1 - R_j^2 \tag{1}$$

$$VIF = \left[\frac{1}{Tolerance}\right] \tag{2}$$

where: R_j^2 is the coefficient of determination.

Proposed method

Random forest

RF model, introduced by Breiman in 2001, is a powerful method for classification, regression, and unsupervised learning tasks, recognised for its improved predictive performance in various domains (Ait Naceur et al. 2022). It builds a set of decision trees on random subsets of the training data via bagging. The RF evaluates the prediction error by permuting the values of variables on observations not used for tree training. This model is appreciated for its ability to reduce overfitting while offering high accuracy and low errors. Performance is optimised by three key hyperparameters: the number of trees (ntree), the maximum number of features at each division (mtry), and the minimum size of terminal leaves (nodesize) (Sharma et al. 2021). In this study, the Random Forest (RF) model was trained using the full set

of causal factors prepared for landslide susceptibility assessment. The number of trees (ntree) was set to 100, ensuring both model stability and computational efficiency. The mtry parameter was set to 4, following a parameter tuning process that tested multiple values (3, 4, 6, and all features) and selected the value minimising the Out-of-Bag (OOB) error. The model was trained using bootstrap aggregation, whereby roughly one-third of the samples are excluded from each tree and used as OOB data for internal validation. The resulting OOB score was 0.9899, indicating excellent model generalisation. Additionally, a 10-fold stratified cross-validation yielded a mean accuracy of 0.9903 with a standard deviation of 0.0022, demonstrating the stability and robustness of the model across different data partitions. The close agreement between the OOB score and cross-validation results confirms the absence of significant overfitting and supports the reliability of the model for landslide susceptibility prediction.

Multivariate adaptive regression spline

MARS model, developed by Friedman in 1991, is a powerful non-parametric approach for modelling nonlinear relationships between independent and dependent variables although providing an explainable model (Javidan et al. 2019). MARS utilises a regression-based algorithm to create a flexible model that can capture complex relationships. The model consists of a regression function defined as the sum of weighted basis functions (splines), where each spline is represented by a basis function modified according to knot points and interaction order (Hai et al. 2023). The general equation of MARS is given by Equation 3 (Liao and Liao 2020):

$$f(x) = \beta_0 + \sum_{m=1}^{M} \beta_m \lambda_m(X) \tag{3}$$

where β_0 is a constant, M is the number of basic functions, each of them formed by a coefficient β_m and $\lambda_m(X)$ is an individual basis function or a product of two or more basis functions.

Naïve bayes tree

NB classifier is a classification method based on Bayes' theorem, relying on the assumption of conditional independence among attributes, meaning that attributes are presumed to be entirely independent given the class label (Tien Bui et al. 2012). This approach is advantageous due to its simplicity in construction and the absence of complex iterative parameter estimation schemes. Moreover, the NB model is robust to noise and irrelevant attributes, demonstrating its effectiveness across various application domains (Soria et al. 2011).

The modelling of NBT begins by constructing a tree using a feature selection method based on entropy, which measures the level of disorder. Let S represent the set of training data, with $|S|$ being the total number of conditioning factors. These factors are divided into n distinct classes Si ($i = 1,2,\ldots,n$). The number of conditioning factors in each class Si is denoted by $|Si|$. The classification of S is then determined by calculating the expected entropy, as shown in Equation 4 (Nhu et al. 2020):

$$Entropy(S) = -\sum_{i=1}^{n} \left(\frac{|S_i|}{|S|}\right) \log_2 \left[\left(\frac{|S_i|}{|S|}\right)\right] \tag{4}$$

Consider attribute A, for example, aspect, in set S. The expected entropy can be expressed as (Equation 5):

$$Entropy_A(S) = -\sum_{i=1}^{n} \left(\frac{|S_i|}{|S|}\right) \times Info(S_i) \tag{5}$$

The difference between *Entropy(S)* and *EntropyA(S)* is represented as the information gain (InfoGain) (Equation 6):

$$\text{InfoGain}(A) = Entropy(S) - \text{EntropyA}(S) \tag{6}$$

The information gain ratio (IGR) is calculated according to the following equation (Equation 7):

$$GainRatio\,(A) = \frac{\text{InfoGain}\,(\text{A})}{SplitInfo\,(A)} = \frac{Entropy\,(S) - \text{EntropyA}\,(\text{S})}{-\sum_{i=1}^{n}\left(\frac{|S_i|}{|S|}\right)\log_2\left[\left(\frac{|S_i|}{|S|}\right)\right]} \tag{7}$$

Boosted regression tree

BRT is a powerful non-parametric machine learning method, particularly beneficial in the field of data mining for classification and prediction tasks. According to Elith et al. (2008), BRT assumes no pre-established relationship between independent and dependent variables, enabling it to efficiently identify and evaluate relevant predictors. James et al. (2013) states that the hierarchical structure of regression trees suggests that the impact of a variable is influenced by the values of nodes higher up the tree, implicitly representing the relationships between variables.

While BRT is useful for identifying key predictors and understanding their influence, it can be of little use if the analysis focuses too much on minor variations in the data, which could hamper continuous improvement efforts. Nevertheless, the boosting technique can improve the accuracy of decision tree predictions by combining several weak learners into a stronger predictor. The modelling process begins by choosing a splitting variable that best separates the dependent variable, then dividing the results into two classes based on the largest divisor. Recursive division continues until the nodes are established (Saha et al. 2023).

Recursive division continues until the nodes are completely uniform, considering when division should stop to avoid overlearning, as overly complex trees can become unstable and error-prone (Therneau et al. 2015).

Model performance evaluation

The performance of the three models in this study was assessed using the area under the receiver operating characteristic curve (AUC) along with standard statistical metrics: sensitivity (recall), specificity, precision, and accuracy (Ait Naceur et al. 2025). Sensitivity measures the proportion of landslide pixels correctly predicted, specificity evaluates the correctly identified non-landslide pixels, precision represents the proportion of true positives among all predicted positives, and accuracy indicates the overall rate of correct predictions (Dou et al. 2023).

We placed particular emphasis on the AUC because it is threshold-independent and provides a robust way to compare the discriminative ability of different models across datasets (Dou et al. 2023; Abbas et al. 2024). While sensitivity, specificity, and precision give insight into specific aspects of model performance, the AUC summarises the overall predictive capacity, offering a clear picture of how well the model distinguishes landslide-prone areas from stable zones (Wang et al. 2015; Youssef et al. 2016). By combining these metrics, we were able to obtain a comprehensive and balanced evaluation of the models' reliability and effectiveness in mapping landslide susceptibility. The equations used to compute these metrics are as follows (Equations 8, 9, 10 and 11):

$$\text{Specificity} = \frac{TN}{FP + TN} \tag{8}$$

$$\text{Precision} = \frac{TP}{FP + TP} \tag{9}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{10}$$

$$\text{Accuracy} = \frac{TP + TN}{FP + TN + TP + FN} \tag{11}$$

Result

Assessment of landslide causatives factors importance

Conditioning factors are critical components in modelling susceptibility to ground movements because they give information about the underlying circumstances and triggering processes of these events. The RF approach was utilised in this study to determine the relative importance of each component since it can evaluate complex datasets while providing robust and reliable estimates of variable influences. RF is distinguished for its capacity to integrate many explanatory factors while finding non-linear interactions between them. Nonetheless, it is necessary to eliminate components with zero or marginal predictive value to maximise model performance and reduce noise in findings (Naceur et al. 2022). This method not only increases the relevance of the models, but it also concentrates the study on the factors that are most relevant in forecasting ground motions, resulting in more precise and rigorous modelling.

Analysis of the results obtained in our present study reveals that slope angle, elevation, and distance from streams are the main factors influencing susceptibility to ground movement. Slope angle stands out as the most influential factor, with a significance of 0.171. This result is consistent with the literature, as steep slopes increase the risk of gravitational instability. Elevation, with a significance of 0.152, also plays a significant role, influencing hydrological gradients and erosion processes that can affect slope stability. Distance to streams, with a value of 0.144, occupies an important place. This factor reflects the impact of fluvial erosion and water accumulation on adjacent slopes, which can lead to instability. Distance to roads (0.133) and lithology (0.113) follow closely in terms of importance. Proximity to road infrastructures is often associated with local geotechnical modifications, while lithology determines the mechanical strength of soils and rocks. The NDVI, with a significance of 0.09, demonstrates that vegetation cover protects against erosion. TWI is significant at roughly 0.08, indicating the relevance of soil moisture in slope stability. Other parameters, such as TPI and slope aspect, have a moderate effect (0.027). Finally, land use (0.023), precipitation (0.019), and curvature (0.016) appear to be the least relevant elements in our investigation, most likely due to their low geographic variability or poor association with land movement events in the studied region (Figure 5).

Multicollinearity analysis

The results of the multicollinearity analysis show that the factors influencing susceptibility to ground movement exhibit acceptable levels of collinearity (Figure 6). Tolerance values (TOL) range from 0.365 to

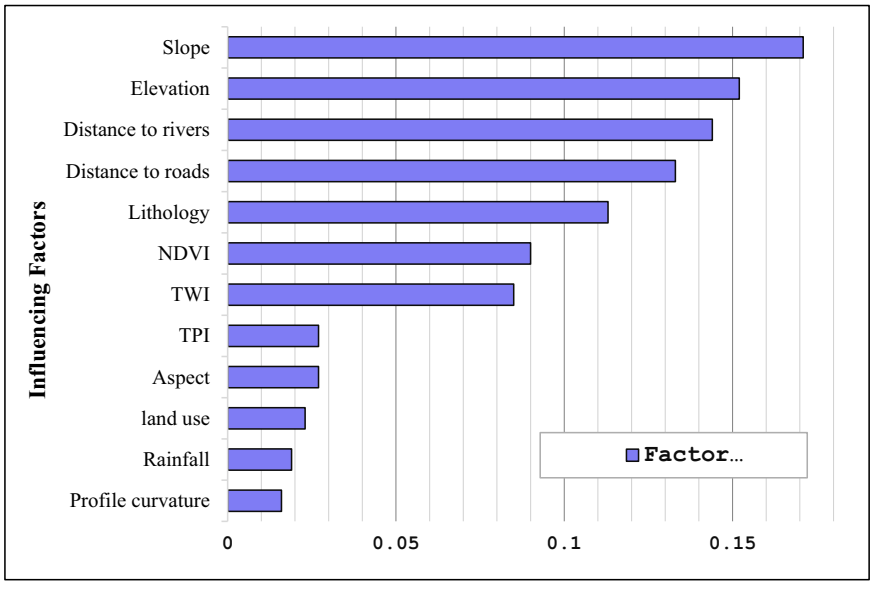


Figure 5. Relative importance of the conditioning factors.

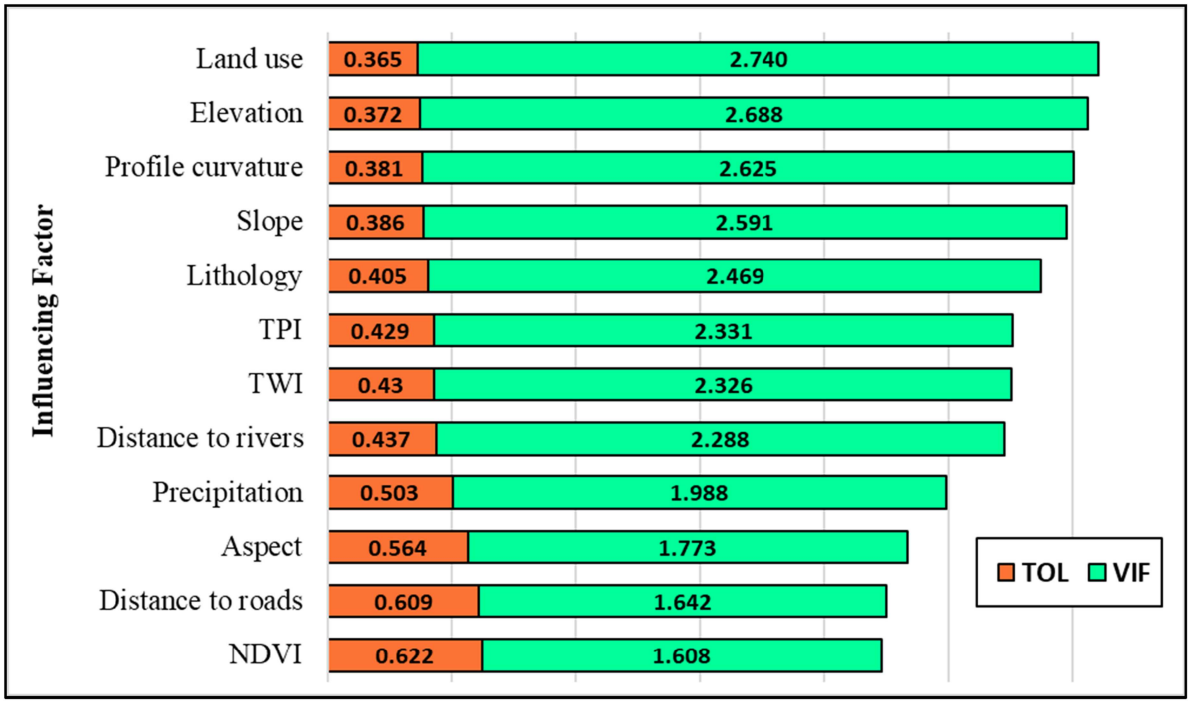


Figure 6. Multicollinearity analysis in the current study.

0.622, indicating that no factor approaches a critical multicollinearity value (generally < 0.1). In addition, the factors show Variance Inflation Factor (VIF) values between 1.608 and 2.740, well below the critical threshold of 10, confirming the absence of severe collinearity. Variables such as land use (VIF = 2.740), and elevation (VIF = 2.688) have the highest VIF values but remain within acceptable limits. These results suggest that the factors can be reliably used for modelling without the risk of coefficient distortion due to multicollinearity. While the multicollinearity analysis confirms that all factors are within acceptable limits, performing a sensitivity analysis could further clarify the influence of variable selection and parameter tuning on model outputs.

Comparative analysis of model performance

The performance of the RF-MARS, RF-BRT, and RF-NBT models was evaluated using ROC curves for both the training and validation datasets (Figure 7). A consistent and independent 70/30 training–validation split was applied across all models to ensure fair comparison and to assess model generalisation. Analysis of the training data (Figure 7A) shows AUC values of 93.66%, 91.96%, and 91.10% for the RF-MARS, RF-BRT, and RF-NBT models, respectively, indicating strong performance across all models, with a slight advantage for RF-MARS. For the validation data (Figure 7B), the AUC values are 92.73% for RF-MARS, 91.48% for RF-BRT, and 90.71% for RF-NBT. The limited discrepancy between training and validation AUC values suggests that overfitting is minimal and that the models exhibit stable predictive behaviour. These results demonstrate the stability and reliability of the developed models, with RF-MARS consistently outperforming the others in both training and validation phases.

Additionally, four statistical indices, namely recall, specificity, accuracy, and precision, were also used to assess the performance of the models applied in this study (Figure 8). All metrics were computed using the same independent validation dataset to ensure consistency in performance assessment. The RF-MARS model shows the best results across all indicators, with a recall of 93.12%, specificity of 94.19%, accuracy of 93.65%, and precision of 94.26%. These values suggest that the RF-MARS model is particularly effective in accurately identifying landslide-susceptible areas while minimising false alerts. The RF-BRT model ranks second, with recall, specificity, accuracy, and precision values reaching 91.13%, 92.50%, 91.80%, and

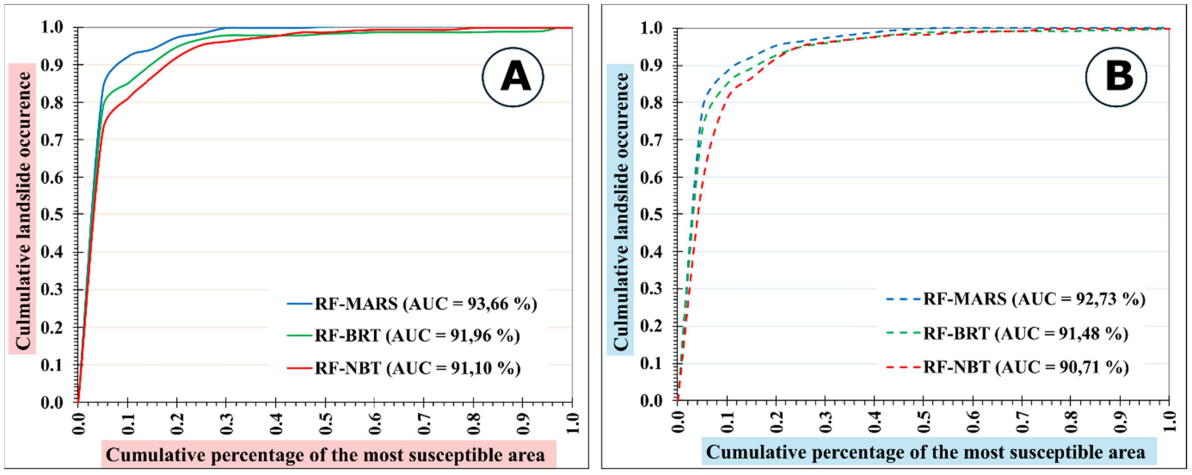


Figure 7. ROC curves and corresponding AUC values illustrating the performance of the models: (A) using the training dataset and (B) using the validation dataset.

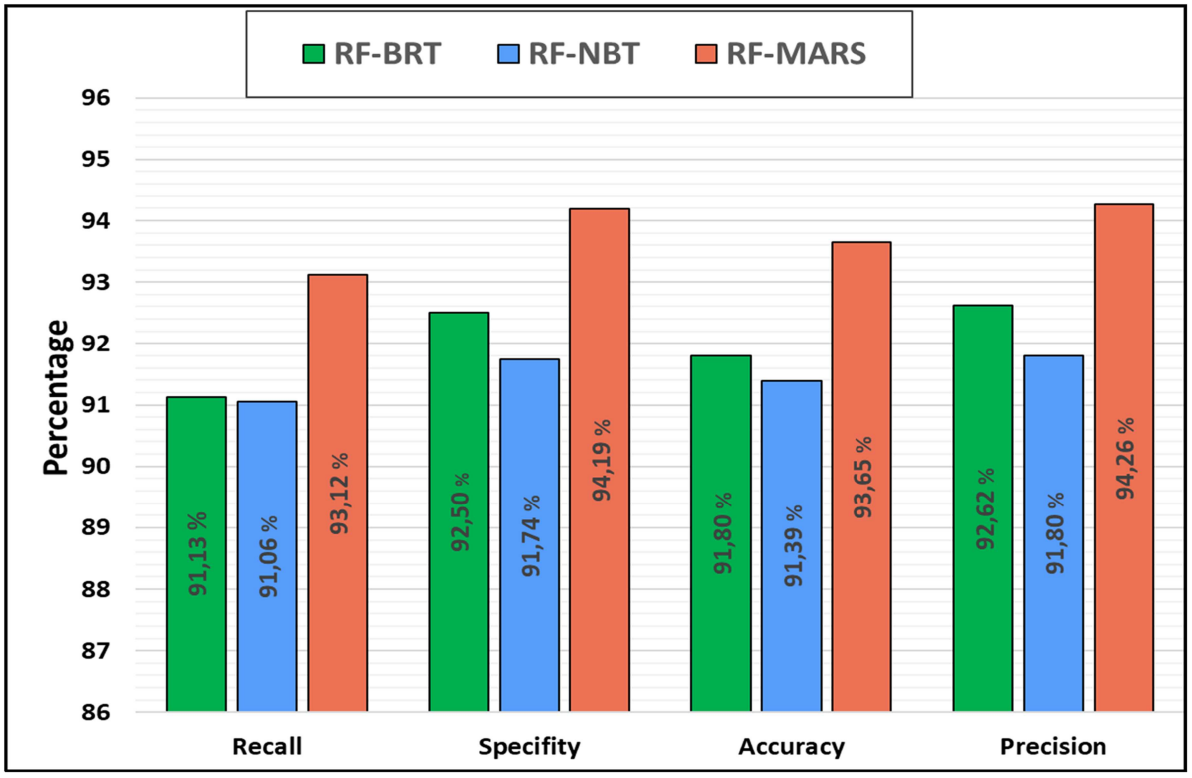


Figure 8. Statistical validation metrics of the three applied models.

92.62%, respectively. Although its performance is slightly lower than that of RF-MARS, RF-BRT remains a reliable and effective model for landslide susceptibility mapping. In contrast, the RF-NBT model shows slightly lower values, with a recall of 91.06%, specificity of 91.74%, accuracy of 91.39%, and precision of 91.80%, placing it third in terms of performance. These results underscore the robustness of the RF-MARS model in predicting landslide-prone areas, highlighting its suitability for risk management applications. Although a single independent 70/30 training–validation split was adopted, the close agreement between training and validation results indicates minimal overfitting. Future studies may include k-fold cross-validation or external datasets to further enhance model generalisability.

Landslide susceptibility mapping

The RF-MARS, RF-BRT, and RF-NBT hybrid models were used to generate landslide susceptibility maps, combining the advantages of each approach. We achieved model hybridisation by using the RF model's results as input data for other algorithms, thereby maximising prediction accuracy. In this study, the models were trained in two key steps. The first step was to calculate a susceptibility index for each pixel in the study area, considering the causative factors identified. At this stage, each pixel is assigned a unique susceptibility value calculated from the selected hybrid algorithm. Then, in the second stage, the susceptibility indices generated were reclassified into distinct categories using the geometric interval method, which enables optimal classification of continuous data (Pham et al. 2017). This method, recognised for its ability to reduce intra-class variance, enabled the susceptibility indices to be subdivided into five risk classes: very low, low, moderate, high, and very high. This classification was used to produce accurate susceptibility maps, illustrating the spatial distribution of areas potentially prone to landslides (Figure 9).

Landslide susceptibility maps, generated in the present study, reveal distinct spatial distributions across the Oued Lakhdar watershed. The map obtained with the FR-MARS model shows a significant concentration of moderate-to-high susceptibility areas, mainly in the southern and central mountainous zones. Areas of very low susceptibility are mainly located in sectors with flatter topography and lower elevations. In contrast, the map based on the FR-NBT model shows a decrease in areas of very high susceptibility while enlarging areas of moderate susceptibility, indicating a broader detection of areas at risk. Low-susceptibility zones are more widespread in the central and northern parts of the watershed.

The FR-BRT model, on the other hand, takes a more conservative approach, with areas of very high susceptibility limited to sectors with specific geomorphological features, notably the steep slopes to the south. Zones of low to very low susceptibility cover a large part of the map, particularly in low-lying areas. The hybrid approach exploiting the results of the RF algorithm as input data thus maximises the efficiency

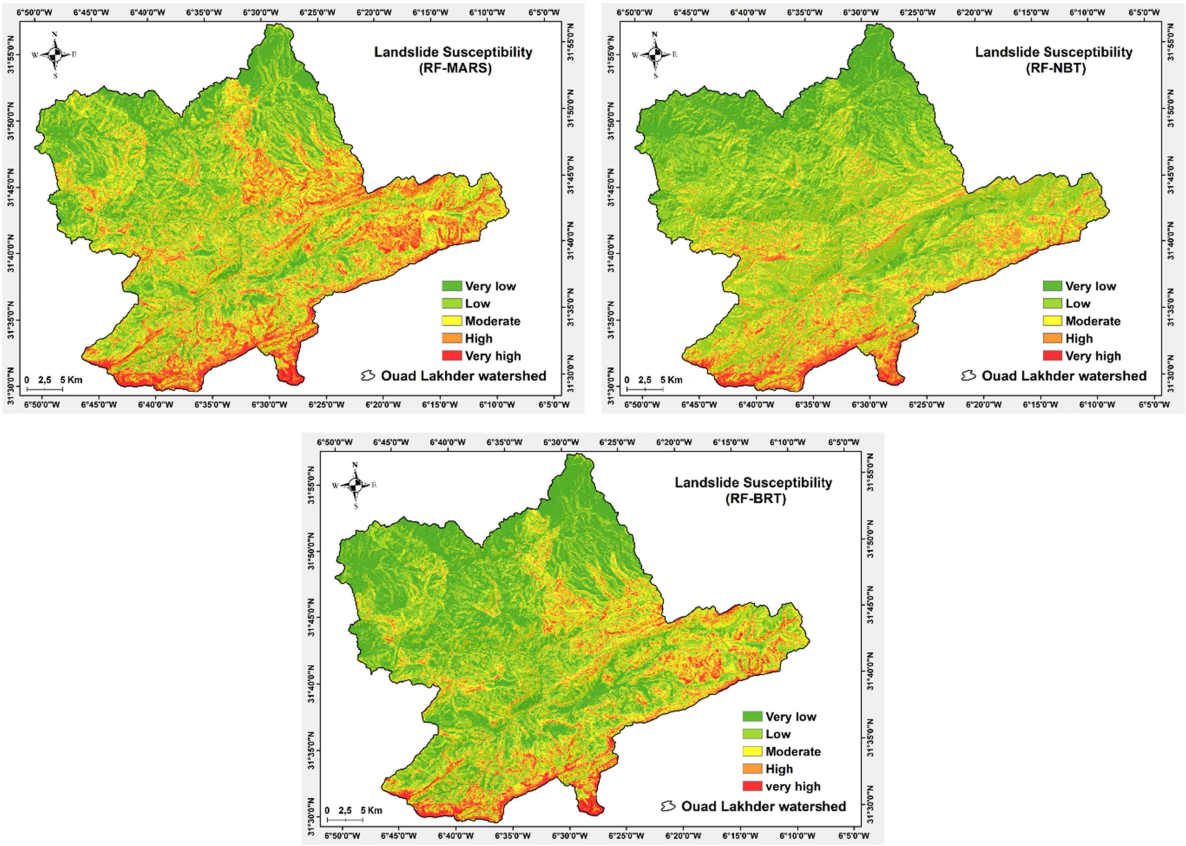


Figure 9. Spatial assessment of landslide susceptibility in the Oued Lakhdar Basin through three models.

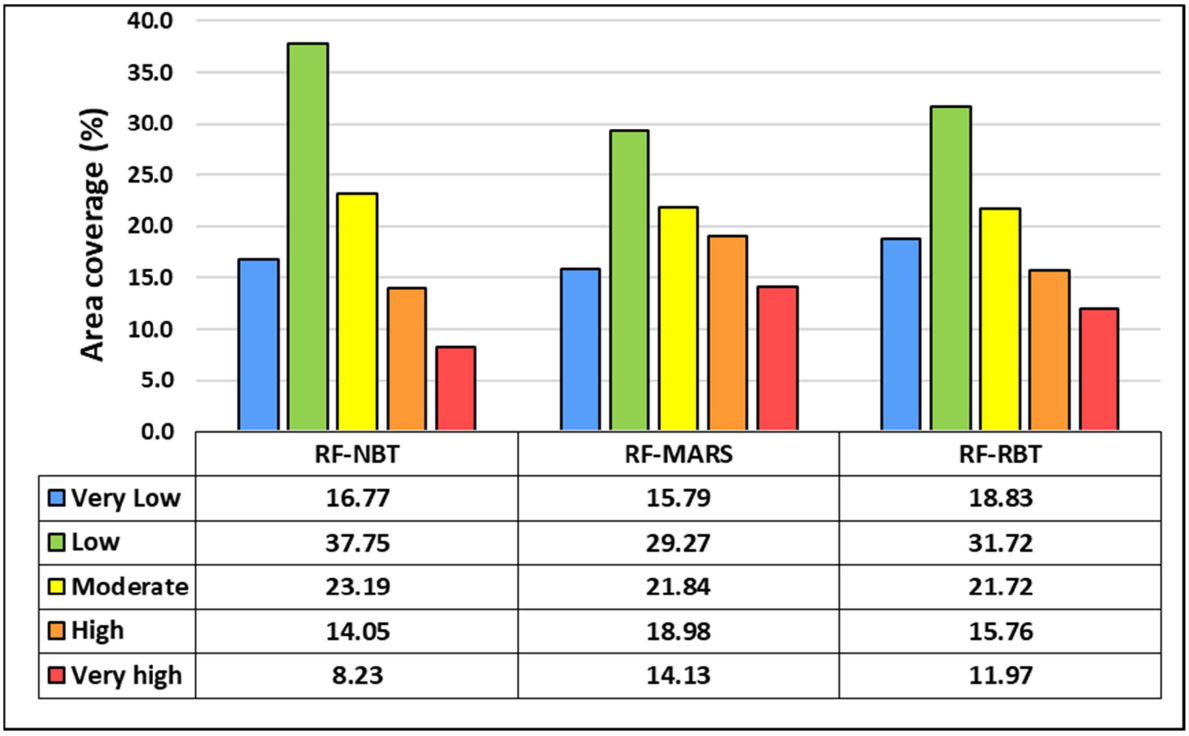
of each model while providing a comprehensive visualisation of risk. The differences between the maps underline the importance of selecting a model suited to the specific objectives of landslide risk analysis and management.

The percentage results for the different susceptibility classes of the three RF-NBT, RF-MARS, and RF-BRT models reveal interesting variations in the spatial distribution of risk zones. Overall, the three models show different proportions for each susceptibility class, underscoring the differences in sensitivity between the algorithms.

The RF-NBT model shows the highest proportion of the low susceptibility class with a percentage of 37.75%, followed by the moderate (23.19%), very low (16.77%), high (14.05%), and very high (8.23%) classes. This suggests that this model tends to be more conservative, classifying a large part of the study area into the low to moderate risk categories. The RF-MARS model, on the other hand, exhibits a different distribution, with a higher proportion of areas classed as high susceptibility (18.98%) and very high susceptibility (14.13%), while a lower proportion are classified as low (29.27%) and very low (15.79%). This suggests that the model sees a larger portion of the area as more prone to landslides. Finally, the RF-BRT model produces an intermediate distribution with roughly equal proportions amongst classes. It identifies 31.72% of the area as having low susceptibility, 21.72% as moderate, and 18.83% as extremely low. Furthermore, 15.76% of the area is classified as high vulnerability, while 11.97% have very high susceptibility (Figure 10).

Discussion

In this study, the landslide susceptibility of the Oued Lakhdar watershed was assessed using three hybrid machine learning models: RF–NBT, RF–BRT, and RF–MARS and their predictive performances were systematically compared with previous studies. The RF–NBT model yielded AUC values of 91.10% for training and 90.71% for validation, indicating strong predictive power with minimal overfitting. These results are consistent with prior investigations on Naïve Bayes Tree-based approaches; for instance, Chen et al. (2017) reported an AUC of 0.880 for NBTree, and 0.883 for a hybrid Rotation Forest–NBTree approach in Langao County, China, underscoring the accuracy gains provided by ensemble strategies.



	RF-NBT	RF-MARS	RF-RBT
Very Low	16.77	15.79	18.83
Low	37.75	29.27	31.72
Moderate	23.19	21.84	21.72
High	14.05	18.98	15.76
Very high	8.23	14.13	11.97

Figure 10. Proportion of landslide susceptibility classes obtained from the three models.

Similarly, Arabameri et al. (2021) observed AUCs of 0.759 and 0.728 for training and validation with NBTree alone, showing that even standalone models can reliably identify landslide-prone areas, though with slightly lower predictive power.

For the RF–BRT model, the present study achieved an AUC of 91.96% for training and 91.48% for validation, complemented by high precision (92.62%), recall (91.13%), specificity (92.50%), and overall accuracy (91.80%). These metrics align well with previous BRT-based applications; Youssef et al. (2016) reported AUCs of 0.958 and 0.856 for success and prediction rates in Wadi Tayyah Basin, while Saha et al. (2021) applied an ensemble CP–BRT model in Rudraprayag, India, achieving cross-validated AUCs of 86.673% and 89.460% with precision between 0.790 and 0.829, highlighting the improved reliability of ensemble methods. Notably, the metrics obtained in the present study slightly exceed those reported in earlier works, reflecting the model's robustness in capturing the spatial distribution of landslide susceptibility and minimising misclassification.

Regarding the RF–MARS model, our results showed exceptional performance with AUC values of 93.66% and 92.73% for training and validation, and high recall, specificity, accuracy, and precision (93.12%, 94.19%, 93.65%, and 94.26%, respectively). When compared to Wang et al. (2015), who reported AUCs of 0.81 and 0.79 using a MARSpline model in Mizunami City, Japan, the RF–MARS hybrid approach clearly enhances both AUC and precision metrics, likely due to the integration of Random Forest for factor selection and weighting, which strengthens spatial prediction reliability. Overall, these results demonstrate that hybrid ensemble models provide superior landslide susceptibility predictions, particularly in semi-arid environments, while highlighting the importance of careful calibration, relevant factor selection, and high-quality input data for achieving robust and accurate predictions.

Conclusion

This study emphasises the relevance of landslide susceptibility mapping in semi-arid regions, where complex geomorphological conditions and increasing anthropogenic pressure contribute to the amplification of landslide hazards. By integrating geospatial datasets with machine learning techniques, a hybrid modelling framework was developed using a comprehensive inventory of 249 landslides and twelve conditioning factors, including essential geomorphological parameters such as slope, aspect, elevation, curvature, topographic wetness index (TWI), and terrain position index (TPI). These factors collectively capture the spatial variability of landslide-prone environments.

The comparative analysis of the tested models demonstrates that the RF–MARS approach achieved the highest predictive capability, with an AUC of 93.66% during the validation phase. This performance highlights the robustness of the hybrid model and its effectiveness in capturing the non-linear interactions between environmental factors and landslide occurrence in semi-arid mountainous landscapes.

The landslide susceptibility maps generated in this research provide valuable spatial information for hazard mitigation and territorial planning. While the maps identify areas with varying degrees of susceptibility, it should be noted that the predictions are probabilistic, and the associated uncertainty should be considered when interpreting the results. The susceptibility classes were defined using the Natural Breaks method, which optimises the classification boundaries based on the data distribution. Decision-makers can use this information to prioritise attention to areas with higher predicted susceptibility, while recognising the inherent limitations of the model.

Beyond the specific context of the Oued Lakhdar watershed, the proposed methodological framework is transferable and can be applied to other semi-arid regions exhibiting comparable geomorphological and climatic settings. Nevertheless, some limitations should be acknowledged. The integration of higher-resolution datasets, additional conditioning variables, and emerging machine learning or deep learning techniques could further enhance predictive accuracy and spatial generalisation. Overall, this study contributes to the growing body of research on landslide susceptibility assessment and represents a meaningful step toward more informed, data-driven, and sustainable landslide risk management in semi-arid environments.

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Author contributions

CRediT: **Hassan Ait Naceur:** Conceptualization, Data curation, Resources, Writing – original draft; **Brahim Igmoullan:** Data curation, Investigation, Writing – original draft; **Mustapha Namous:** Methodology, Validation, Writing – review & editing; **Youssef Lassiane:** Investigation, Writing – original draft, Writing – review & editing; **Hoda Benazun:** Methodology, Validation, Writing – review & editing; **Javed Mallick:** Investigation, Writing – original draft, Writing – review & editing; **Hoang Thi Hang:** Methodology, Writing – review & editing; **Hazem Ghassan Abdo:** Investigation, Writing – review & editing.

Disclosure statement

All authors declare that they have no conflict of interest.

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Data availability statement

The data that support the findings of this study are available on request from the corresponding author.

Ethics approval

This article does not contain any studies with human participants or animals performed by any of the authors.

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